

THIS PCB's PURPOSE or DESIGN GOAL:

- * update MITS/Altair CPU-A board for modern EMC PCB layout principles
 - ** power plane(s) to reduce Vcc and Ground shifting niose
 - ** place decoupling capacitors next their respective ICs
- * improve processing speed above 1975 technology limited 8080A with newer CPU
 - ** HMOS 8085A upto 6MHz (late 1970s)
 - ** CMOS 80C85B upto 8MHz (mid 1990s)
- * add low voltage or 5Vcc < 4.7V forced RESET to help power up reliability
- * same PCB to support replacement of IMSAI 8080A MPU-A
 - ** have 2 ribbon cable connectors
 - ** S-100 jumpers and circuits for supporting signal varations
- * add Power On Jump to a user defined address (1K increments)
- * minimal hardware, no IO, no RAM, no ROM, etc... just CPU
- * allow for easy changing of CPU MHz speed
 - ** use DIP oscillator as flexible source
 - ** have PCB holes to allow HC case quartz crystal
- * design a spin off varation to use the NSC800 (a Z80A hybrid with 8085 like pin out)

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Top Sheet, Repro update
Altair CPU-A with 8085A/80C85B

Sheet: /
File: CPU-85_r104.kicad_sch

Title:

Size: A4

Date: 6 August 2026, greglori_mi@wowway.com

Rev: rev 1.04

KiCad E.D.A. 10.0.4

Id: 1/12

CHANGE HISTORY:

- 6/13/2026 thru 7/4/2026 rev 1.0
- * started to make from IMSAI MPU-B rev 3.43 schematic by deleting parts
 - * made new unique oscillator foot print to allow use of either 8 or 14 pin DIP package
 - * changed all but 1 twin 2 pin jumpers (like J2 and J3) to 3 pin jumper
 - * redesigned CPU state outputs to Front Panel based on MPU-A, CPU-A and MPU-B schematics
 - * added 8 pin Altair Front Panel connector to support both IMSAI and MITS/Altair applications
 - * added page 7 Power On Jump feature
 - * 4 layer Gerbers made, no routing or DRC issues
 - * no PCBs made
- 7/4/2026 rev 1.01
- * changed PCB to a 2 layer PCB
 - * 2 layer Gerbers made, no routing or DRC issues
 - * no PCBs made
- 7/8/2026 rev 1.02
- * added C26 for slower CPU crystals
 - * 2 layer Gerbers made, no routing or DRC issues
 - * changed 30u" (Military/Medical grade) to 20u" (Industrial grade) Hard Gold on S-100 fingers
 - * schematic presented to S-100 Vintage Computer Group for review and comments
 - * no PCBs made
- 7/9/2026 to 7/13/2026 rev 1.03
- * add CPU State Latch for driving CPU status sM1 on S-100 Bus DataOut5 for Front Panels
 - * made new PSYNC signal, 3 selectable sources: IMSAI MPU-B ALE as PSYNC made a PSYNC 25% shorter; or M1 from CPU State Decoder; or a pulse lengthened to the end of T2 cycle
 - * renumbered U23 thru U26, made space for added Tri State driver latch U23
 - * two 8T97 changed to 74HCT367, worked OK on existing IMSAI MPU-B at 4MHz
 - * newer KiCAD 10.99 was not compatible with FreeRouter 2.2.4, so had to re-load KiCAD 10.0.4
 - * deleted D5 and R55 POJ indicator LED
 - * FreeRouter 2.2.4 not performing autorouting operation (still investigating on 7/13/2026)
 - * schematic given to S-100 Vintage Computer group for review and to identify issues
 - * no PCBs made
- 7/14/2026 to 8/8/2026 rev 1.04
- * 8080 CPU State Table needs to be implemented not the 8085 CPU State Table in U23
 - * U23 changed from 74HCT138 to 16V8 that can perform some complex gating
 - * feedback sINTA is now a hardwired 16V8 input because WINCUPL did not let me use an 16V8 internal path
 - * restored D5 and R55 POJ indicator LED
 - * auto routing problem with FreeRouter versions between 2.1.0, 2.2.0 and 2.2.4 stalls with unknown problem... Free Router team made aware of the version sensitivity and given the DNS files to help debug this is FreeRouter Bug Report #753
 - * success on 2 layer with Ground plane and 4 layer using older 2.1.0 Free Router
 - * 2 and 4 layer versions PCB layouts made
 - * only 2 layer Gerbers sent out for fabrication as 1.04.2 (18July2026 date in upper left corner)
 - * andrasfuchs emailed issue #753 fixed in upcoming new FreeRouter release v2.3.x(?) (4Aug2026)
- ERRATA for 1.04.2 (corrected for second run with 6Aug2026 date in upper left corner)
- * "T2_LENGTH" label in silkscreen at rev 1.01 J11 panel location, deleted this
 - * DIP switch labels are not on silk screen (lost from early rev 1.01), added these
 - * Vector Interrupt 7 and S-100 finger label 10 too close, looks like "710", moved the "7" label

Reference Specifications:

- Mechanical Outline is from IEEE 696 S-100 Bus Specification 1983-06_Images.pdf
- NOTE1: Bottom S-100 Edge, 45 degree chamfer, 0.60mm into the top and bottom faces, 1/3 depth or 0.6mm of PCB thickness; total PCB thickness 1.60mm to 1.80mm or 0.066inch
- NOTE2: 20u" Hard Gold on S-100 fingers with "lmm" finish
- NOTE3: PCB material is FR4 - 170TG with >200V isolation, UL 94V-0 Flammability
- NOTE4: 2oz copper front and back layers, inner layers 1oz
Green LPI masking, via holes plugged with masking
white silk screen text
- NOTE5: IMSAI MPU-B Schematic Rev 2.pdf as updated by Robret Weatherford 3/2021
- NOTE6: IMSAI MPU-B Reference Manual
- NOTE7: page 24 880-101 CPU-A card schematic from MITS-Altair8800TheoryOperation_1975.pdf
- NOTE8: page 34 of S-100 Bus Handbook by Dave Bursky, Hyaden Book Company, Rochelle Park, NJ
- NOTE9: <https://www.teletext.com/blogs/technical-resources/the-definitive-guide-to-connector-plating-thickness-15%CE%BC-gold-30%CE%BC-gold-and-tin-compared>
- NOTE10: Cromemco ZUP (Z80A) card schematic
- NOTE11: IMSAI Schematics, Page 4 is MPU-A rev4 from 2/76
- NOTE12: Intel MCS80 8080 Microcontroller Systems User's Manual September 1975

SOFTWARE VERSIONS USED:

KiCad

Version: 10.0.4, release build
wxWidgets 3.3.2 Unicode and Boost 1.90.0
Platform: Windows 10 (build 19045), 64-bit edition, 64 bit
GDI objects in use 696

Freerouting - an open-source PCB auto-router

Version: 2.1.0

This program comes with no warranty!

Latest release: <https://github.com/freerouting/freerouting/releases>

Report a problem: <https://github.com/freerouting/freerouting/issues>



S-100 PCB Card Information

Sheet: /Change_History_and_Specs/
File: S100_Card_sch.kicad_sch

Title:

Size: A4

Date: 6 August 2026, greglori_mi@wowway.com

Rev: rev 1.04

KiCad E.D.A. 10.0.4

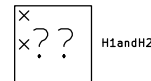
Id: 2/12

J1 S100_MALE_as_2_columns				
1	1	+8_V	+8_V	51
2	2	+16_V	-16_V	52
3	3	XRDY	SSWDSB*	53
4	4	VIO*	EXT_CLR*	54
5	5	VI1*	TMA0*	55
6	6	VI2*	TMA1*	56
7	7	VI3*	TMA2*	57
8	8	VI4*	sXTRQ*	58
9	9	VI5*	A19	59
10	10	VI6*	SIXTN*	60
11	11	VI7*	A20	61
12	12	NMI*	A21	62
13	13	PWRFAIL*	A22	63
14	14	TMA3*	A23	64
15	15	A18	NODEF2	65
16	16	A16	NODEF3	66
17	17	A17	PHANTOM*	67
18	18	SDSB*	MWRT	68
19	19	CDSB*	PROT_STAT*	69
20	20	UNPROT	PROT	70
21	21	DIDSBL*	RUN	71
22	22	ADSB*	RDY	72
23	23	DODSB*	INT*	73
24	24	Phase2	pHOLD*	74
25	25	Phase1	RESET*	75
26	26	pHDLA	pSYNC	76
27	27	pWAIT	pWR*	77
28	28	INTE	pDBIN	78
29	29	A5	A0	79
30	30	A4	A1	80
31	31	A3	A2	81
32	32	A15	A6	82
33	33	A12	A7	83
34	34	A9	A8	84
35	35	D01_BD1	A13	85
36	36	D00_BD0	A14	86
37	37	A10	A11	87
38	38	D04_BD4	D02_BD2	88
39	39	D05_BD5	D03_BD3	89
40	40	D06_BD6	D07_BD7	90
41	41	DI2_BD10	DI4_BD12	91
42	42	DI3_BD11	DI5_BD13	92
43	43	DI7_BD15	DI6_BD14	93
44	44	sM1	DI1_BD9	94
45	45	sOUT	DIO_BD8	95
46	46	sINP	sINTA	96
47	47	sMEMR	sWO*	97
48	48	sHLTA	ERROR*	98
49	49	CLOCK	POC*	99
50	50	OV_GND	OV_GND	100

NOTE: S-100 Labels and PCB Holes Cut & Pasted from
John Monohan's web page https://avitech.com.au/?page_id=3240
(Google Vintage Computer Group, used on IMSAI CPU-A repro panel)
26April2025

That symbol re-imported into KiCad, and the symbol edited
from 1 column of 100 to 2 columns of 50 format to allow fitting
on A sized page; DI, DO and 16 bit pins renamed per page 18 of
IEEE-696 standard; a few pin names corrected ei. IEEE called pins
92 and 93 BiDirectional 3 "even" and BiDirectional 4 "odd"...
which looks to be typo errors to me.
22June2026

Card ejector mounting holes



NOTE: S-100 pin number labels are used to connect to
circuits on these schematic pages.

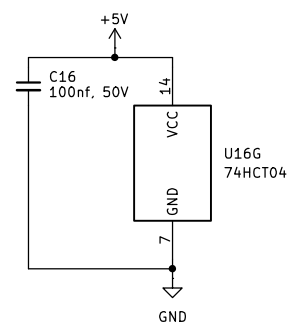
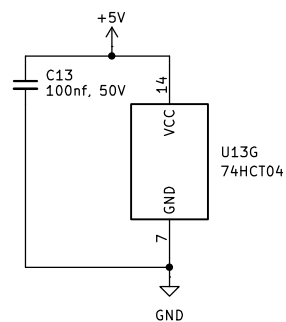
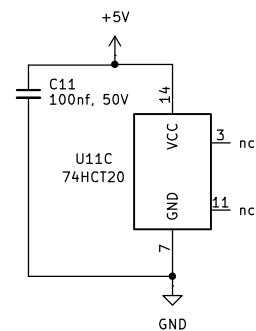
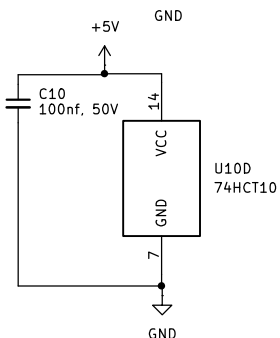
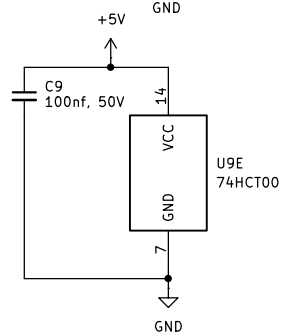
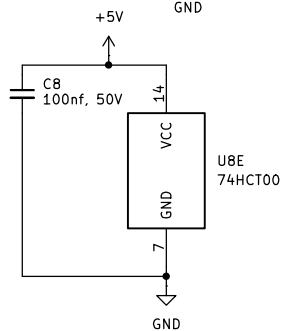
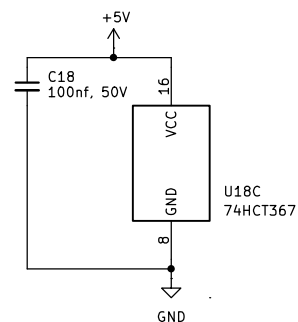
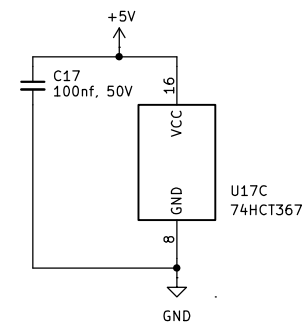
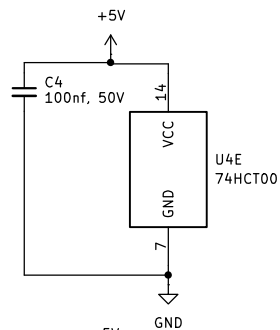
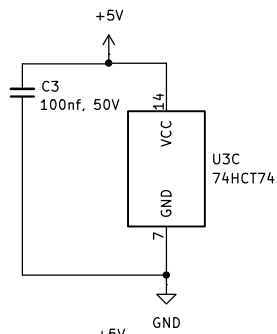
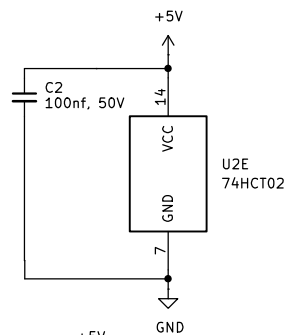
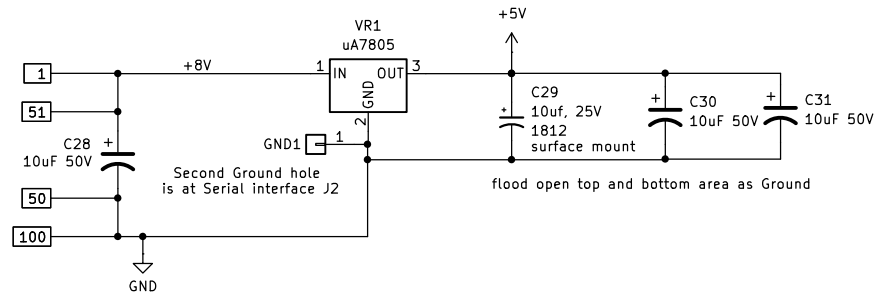
S-100 Connector Assignments

Sheet: /S-100_Bus_Connector/
File: S100_Bus.kicad_sch

Title:

Size: A4	Date: 17 July 2026, greglori_mi@wowway.com	Rev: rev 1.04
KiCad E.D.A. 10.0.4	Id: 3/12	

Note: Use 4 or more vias to connect PCB Top to PCB Bottom copper on +8V and Ground



locate each decoupling capacitor at each respective IC location Vcc pin position

5V Regulator & Decoupling Caps

Sheet: /Power_Supplies/
File: Power_supplies.kicad_sch

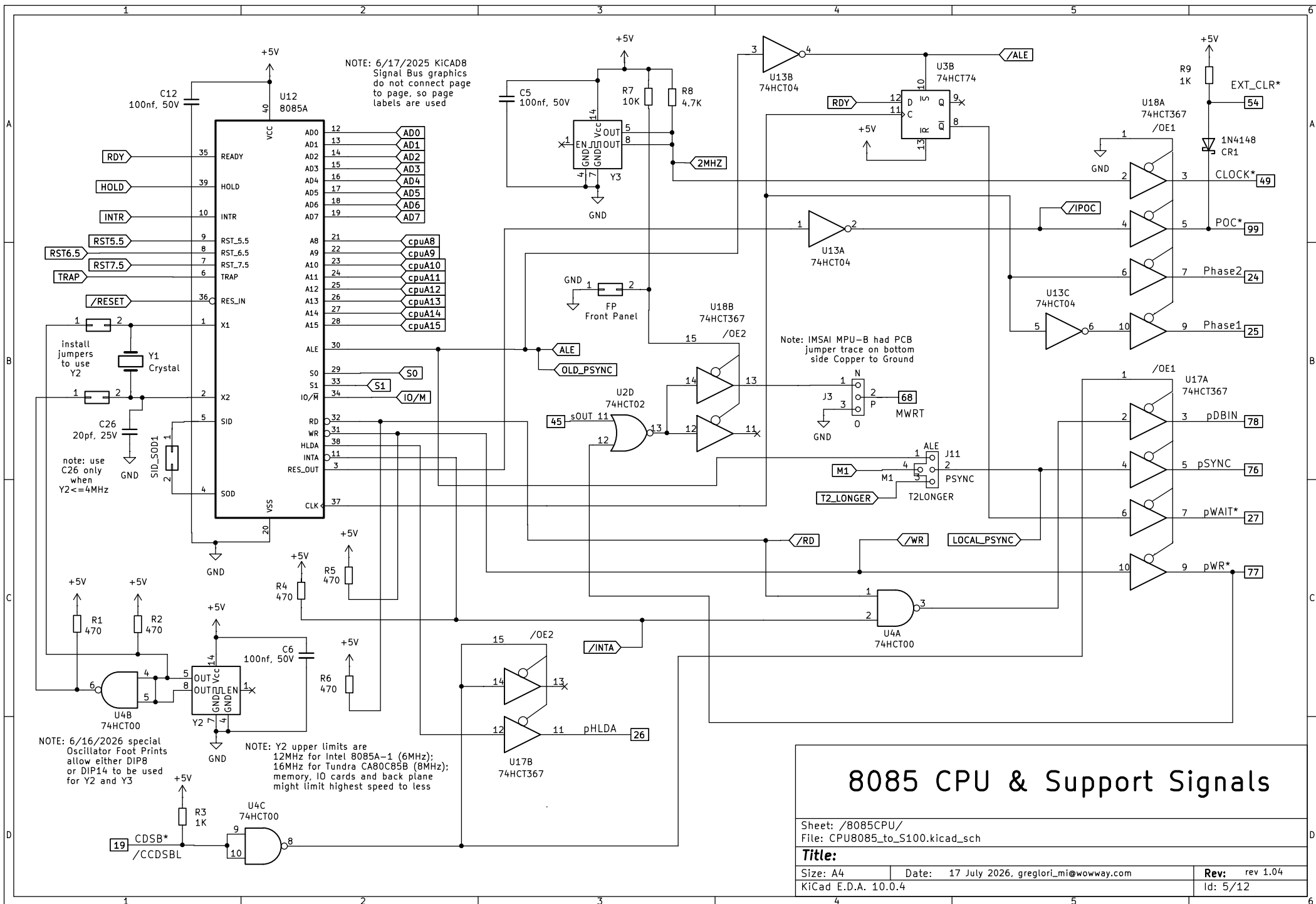
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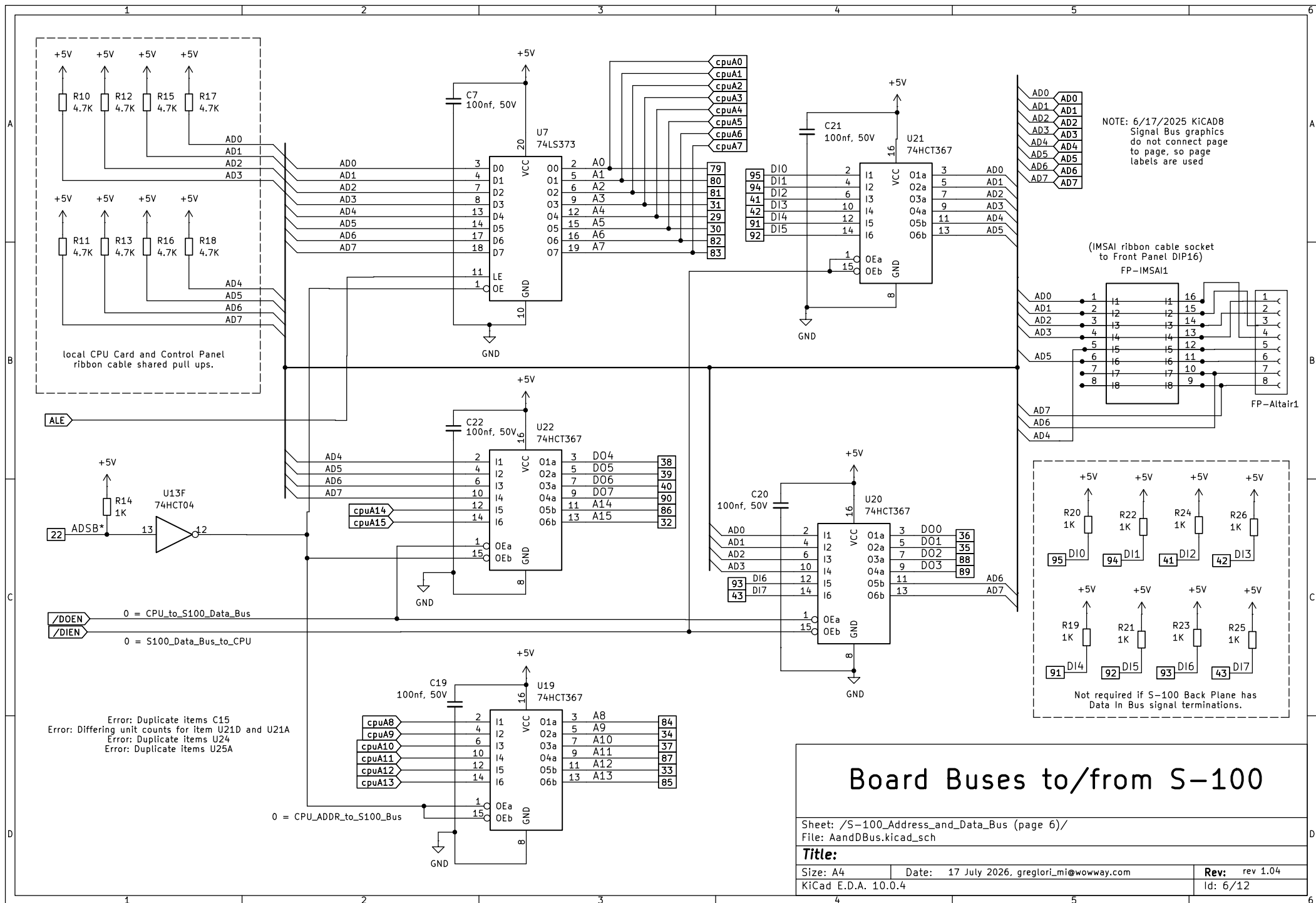
Size: A4 Date: 17 July 2026, greglari_mi@wowway.com

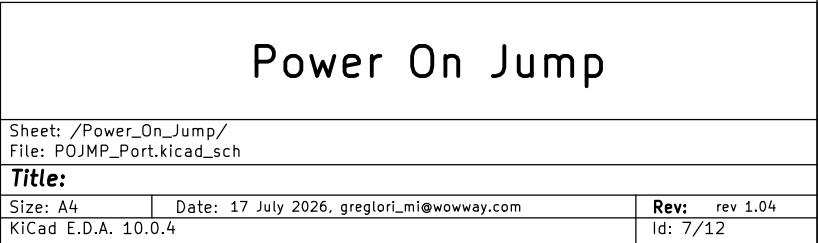
KiCad E.D.A. 10.0.4

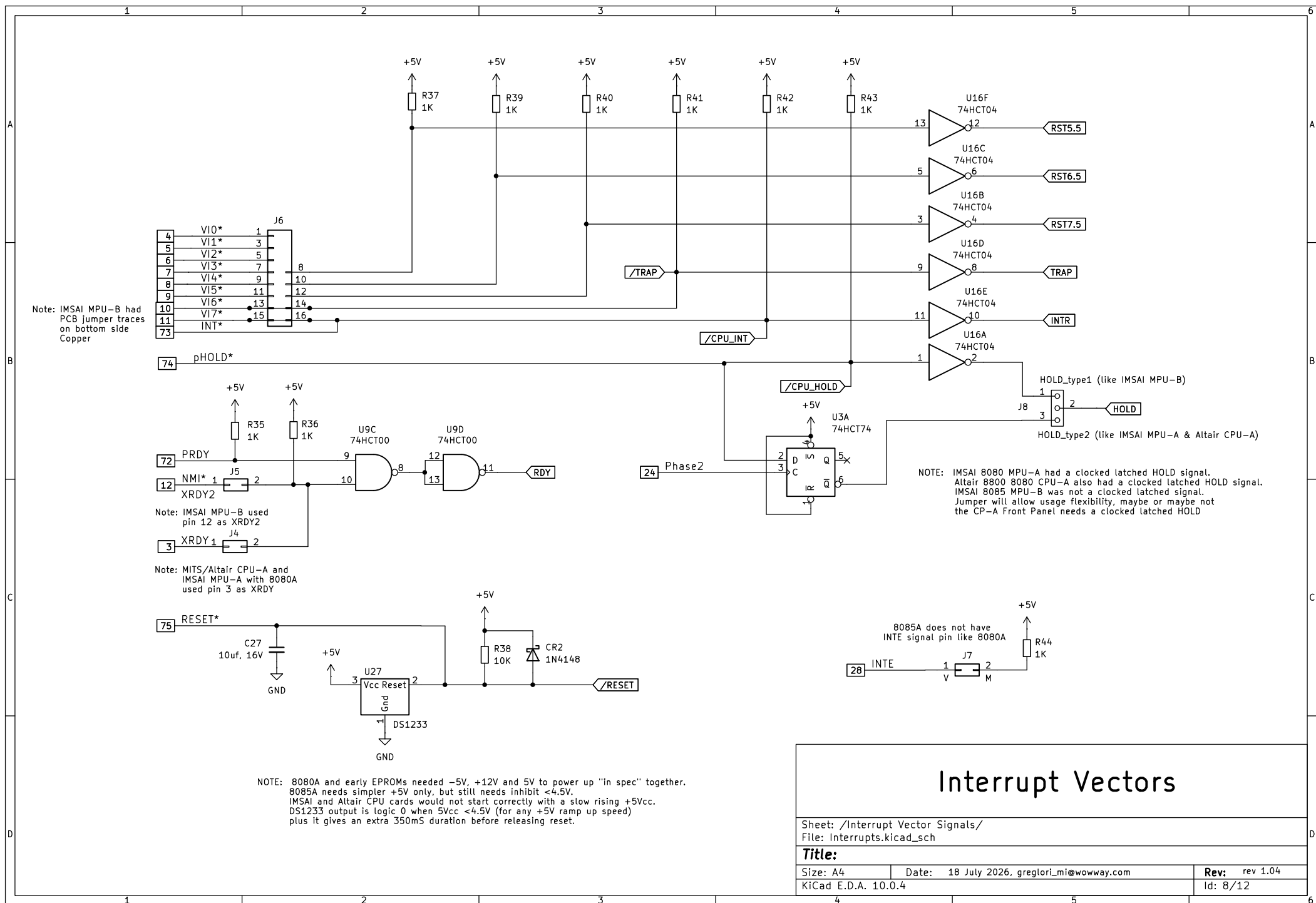
Rev: rev 1.04

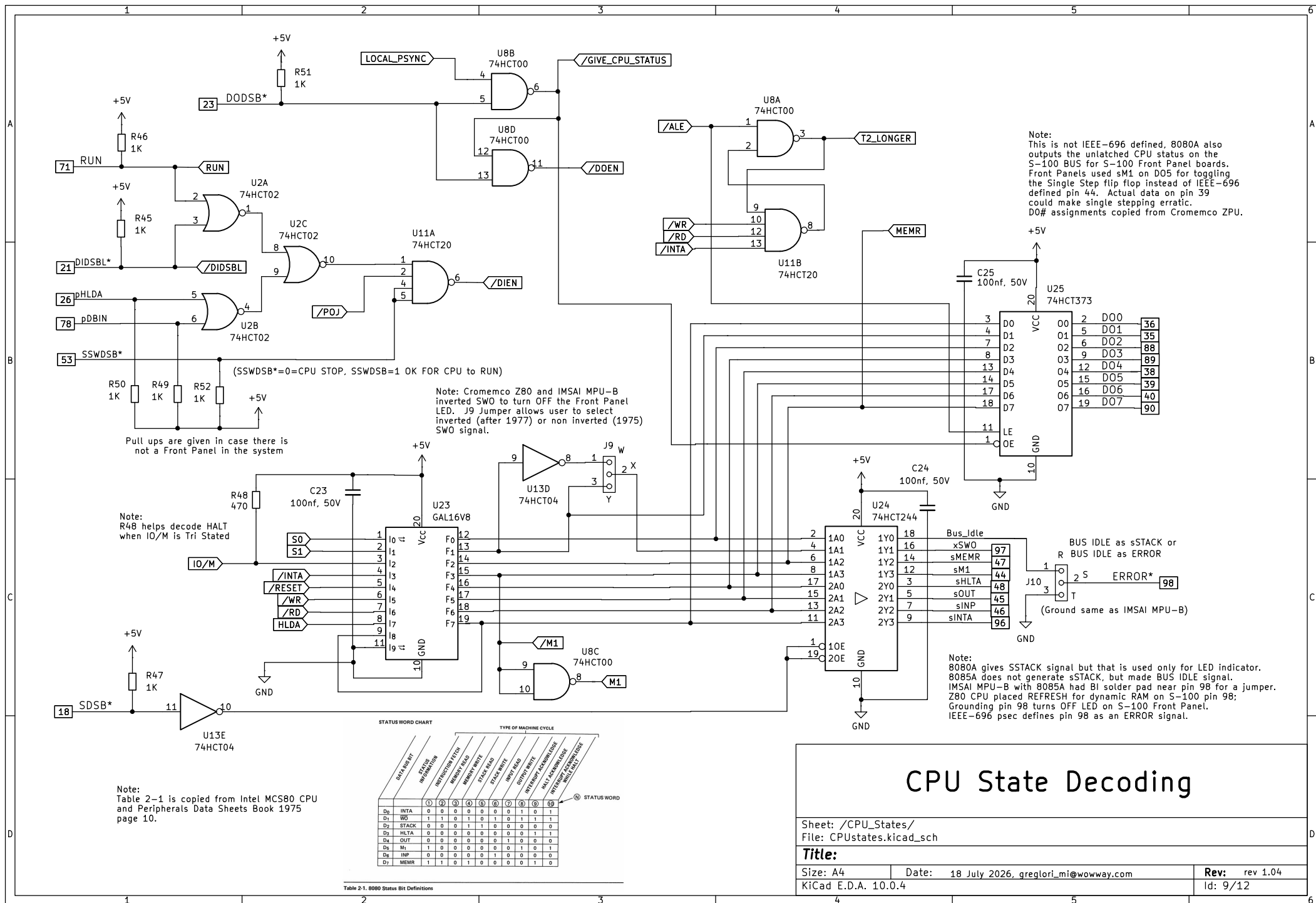
Id: 4/12











ASSEMBLY INSTRUCTIONS:

- 1) solder lowest height parts first... resistors, axial diodes
- 2) then solder DIP sockets, the ceramic capacitors, LED
- 3) and then the 2 and 3 pin SIP headers
- 4) solder tallest parts radial electrolytic capacitors and the VR1 & its optional heat sink last
- 5) use black electrical tape to cover gold electrical edge card fingers and any front to back PCB holes to prevent PCB cleaner seeping under front side parts
- 6) hold PCB vertically and apply about 3 oz, or 1 even coating, over the rear of the PCB of the ZEP Citrus cleaner
- 7) wait about 1 minute
- 8) hold PCB vertically, apply about 3 oz to 4 oz or 1 even washing coat of Kwik Clean Flux Cleaner
- 9) wait about 1 minute
- 10) rinse with distilled water
- 11) shake water off the PCB or use a hair dryer on "Cold" high speed setting to blow dry the PCB
- 12) power up the panel without ICs to check the regulator output voltage (5V)
- 13) without power applied to the panel, install ICs logic, CPU, etc. (Do not "Hot Plug" parts or do not install parts with power active.)
- 14) looking down onto the top of the PCB, look for any outward bent pins that did not go into their socket receivers, remove, fix, un-bend and reinstall parts if necessary
- 15) look from the PCB S-100 edge under all ICs, check for any unseen bent, folded under pins, remove, fix, un-bend and reinstall parts if necessary
- 16) put a drop of rubber cement (not permanent, do not use epoxy) under DIP switch and DIP oscillators to help hold them in place
- 17) set DIP switches and jumper blocks to desired configuration
- 18) power up the CPU-85 and go.... use as needed with your S-100 application(s)

ASSEMBLY INSTRUCTIONS

Sheet: /ASSEMBLY INSTRUCTIONS/ File: Assembly_sch.kicad_sch		
Title:		
Size: A4	Date: 14 July 2026, greglori_mi@wowway.com	Rev: rev 1.04
KiCad E.D.A. 10.0.4		Id: 10/12

CONFIGURATION OPTIONS:

DS1233 +5V Monitor, Low Voltage Inhibit (LVI)
note: almost all CPUs from 1970s lack LVI, MicroChip and Motorola made LVI a standard feature in late 1980s
* do not populate if 1970s performance is desired (not recommended)
(recommend installing to avoid start up and shut down glitching issues on floppy drives)

J11 PSYNC source
* use ALE address latch like IMSAI MPU-B
* use M1 from CPU State Decoder (like Cromemco Z80 board)
* use simulated signal lengthen to last till end of T2 cycle like a 8080A on IMSAI and MITS/ALTAIR boards

install Y2 = maximum allowed by users system

Solder pads for
SID and SOD of 8085

Y1 / Y2 Jumpers
* no jumpers, allows use of Y1 quartz crystal
* per Intel 8085 data sheet page 6 figure 5
* or PDIP oscillator Y2 as CPU clock source
(my preference is for Y2 as DIP oscillator)

Ribbon Cable Interfaces
* 16 pin PDIP for IMSAI Front Panel
* 8 pin Molex for MITS/Altair Front Panel

J8 HOLD - RDY circuit source
* Type1, no latch like IMSAI MPU-B
* Type2, latched like MITS/Altair CPU-A

see Intel 8085 data sheet page 6 Fig. 6

XRDY and XRDY2 sources
* J4 jumper use pin 3 like MITS/Altair
* J5 jumper use pin 12 like IMSAI MPU-B
* can select both as a "wired OR"

J6 Interrupts 5.5, 6.5 and 7.5
note: 8085 contains 3 vectors of a 8214
* jumpers select 3 interrupt sources 1 thru 8

Power On Jump, DIP switches
* switch 1 = ON enables Power On Jump at Reset
* switch 1 = OFF uses CPU default 0000H address
* switch 2, not used
* switch 3 = Address A10
to switch 8 = Address A15.
ON=logic 0, OFF=logic 1
(1K increments from 0100H to FC00H)

J9 xSWO source for S-100 pin 97
* IMSAI MPU-B and Cromemco Inverted xSWO (W-X jumper) turns Front Panel LED default OFF default (a 1977 corrective action)
* or IMSAI MPU-A MITS/Altair xSWO signal (X-Y jumper) as implemented in 1975 but will leave Front Panel LED ON by default (an IMSAI MITS/Altair oops)

J7 INTE S-100 pin 28 (V-M jumper)
note: 8085 does not have an INTE output
* Open Circuit or
* pull up voltage to +5V (like IMSAI MPU-B)

ERROR or SSTACK source for S-100 pin 98
note: 8085 does not make the SSTACK output
* Ground to disable Front Panel LED (S-T jumper) like IMSAI MPU-B
* or decoded BUS IDLE CPU State (R-S jumper)
* MITS/Altair CPU-A and IMSAI MPU-A called this SSTACK
* IEEE-696 defines 98 as bus ERROR signal

FP1, drive source for MRWT S-100 bus signal pin 68
* Open Circuit lets the Front Panel drive MRWT
* If no Front Panel, then install jumper to let CPU drive MRWT this is MRWT Enable (J3 on IMSAI MPU-B)

J3 MRWT Source S-100 pin 68
* CPU driven signal (N-P jumper)
also install FP1 for MRWT Enable
* Ground (P-O jumper)

PCB OPTIONS

Sheet: /CONFIGURATION OPTIONS/
File: Options_sch.kicad_sch

Title:

Size: A4

Date: 5 August 2026, greglari_mi@wowway.com

Rev: rev 1.04

KiCad E.D.A. 10.0.4

Id: 11/12

JUMPER OPTIONS USED/ VALIDATION SET UP:

- 1) Set Power On Jump to go to F800H (for IMSAI PCS Monitor program), OFF for IMSAI and MITS/ALTair Front Panels
- 2) Y1 and C26 not installed, jumpers installed to use Y2=8MHz oscillator until I determine highest feasible MHz
- 3) FP jumper installed to allow CPU to control MWRT, MWRT jumper N-P
- 4) J4 and J5, I installed both jumper blocks to support either MITS/Altair and IMSAI
- 5) J6 no interrupts or jumpers used (user's application specific)
- 6) J7 jumper to have pull up to 5V for INTE (same as IMSAI MPU-B)
- 7) J8 jumper HOLD_TYPE1 closed (same as the IMSAI MPU-B)
- 8) J9 jumper X-Y for not inverting (U24 output is programmed to make the inverted /sW0 for 1977)
- 9) J10 jumpered S-T to Ground (original IMSAI MPU-B setting)

VALIDATION PLANS and RESULTS (RFU, living document):

These procedures are for me, the PCB designer, to validate and verify.

These are FYI for the general user to know the actions I did to check this PCB design (and to be a note to myself to help me know what I did in the future).

- 1) (PASS) KICAD 10.0.4 Design Rule Checking (DRC) tool

```

** Drc report for CPU-85_r2c.kicad_pcb **
** Created on 2026-07-06T12:05:31 **
** Report includes: Errors, Warnings **
** Found 0 DRC violations **
** Found 0 unconnected pads **
** Found 0 Footprint errors **
```
- 2) (PASS) new card must insert and egress with reasonable force from 1 hand into and out of S-100 female receiving connector on S-100 back plane.
- 3) (PASS) new card must slide freely between existing IMSAI S-100 back plane card cage guides.
- 4) (PASS) take 2 blank or production fresh panels, install each into 2 different S-100 female connector manufacturer brands 5 times, visually inspect the female connector's witness marks on the edge connectors for consistent location across all edge pins
(I don't want to wear out my IMSAI back plane's connectors, IEEE spec has a 50 insertion cycle minimum)
black original IMSAI back plane connector:
green Mullen Extended Board:
- 5) (DONE) solder on all parts & sockets after receiving new panels, and install all ICs per Assembly Instructions
- 6) install different Y2 oscillators to find the upper speed limitation of my PCS80/30.
Look for abnormal text on video display (this operation might take 5 minutes per run, enough time for parts to warm up and shift performance if there's a timing issue)
Verify Power On Reset (visually verify LED indicator blinks on Power Up)
2.00 MHz (for CPU speed = 1MHz or XTAL/2)
8.00 MHz
10.00 MHz
11.00 MHz
12.00 MHz
14.00 MHz
15.00 MHz
16.00 MHz
make a PIC12C675 toggle outputs at 0.5Hz into a 80C85B, see if 80C85B can run that slow
- 7) Keep the computer and CPU card powered for 30 minutes with Y2 = highest speed, and take a thermal image to determine if any components are above 45 degrees Celsius or more than 20 degrees above ambient. (higher temperatures indicate abnormal or unexpected loading)
- 8) change jumper from HOLD Type1 to HOLD Type2, and observe if there is an observable difference in operation
- 9) Plug in Don Caprio's repro version of IMSAI Front Panel and observe what features operate and which do not operate.
RESET (Power On Jump Active LED should flash "on" at power up and stay "off")
STOP
RUN
SINGLE STEP
EXAMINE
- 10) ask S100 Vintage Computer Group member with an Altair to try out and return test panel since I don't have a MITS/Altair8800.

VALIDATION PLAN & RESULTS

Sheet: /VALIDATION PLANS/
File: Validation_sch.kicad_sch

Title:

Size: A4	Date: 6 August 2026, greglori_mi@wowway.com	Rev: rev 1.04
KiCad E.D.A. 10.0.4		Id: 12/12